Lecture 13:
Panel data III

PPHA 34600
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From last time: The event study design

In the general version of this model, we include pre-treatment "effects":

$$Y_{it} = \sum_{s=-R}^{S} \tau_s D_i \times \mathbf{1}[\text{periods to treatment} = s]_{it} + \alpha_i + \delta_t + \beta X_{it} + \varepsilon_{it}$$

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- This "lines up" treatment at the same time for everyone
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- We can still use fixed effects to soak up confounders
- We get a (partial) test of the identifying assumption
  - $\rightarrow$ We want pre-treatment τ_s's to be centered on 0 and not trending

Cumulative effects

We're often interested in summing up effects over time:

- This is a simple extension of per-period event study effects
- And very powerful!
- We can use this to understand if treatment moves things in time
- Or happens and fades away
- Or remains important over time

Cumulative effects

We estimate cumulative effects with a distributed lag model:

$$Y_{it} = \sum_{s=0}^{S} \tau_s D_{i,t-s} + \alpha_i + \delta_t + \beta X_{it} + \varepsilon_{it}$$

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- τ_1 gives the marginal effect of treatment 1 period later
- τ_2 gives the marginal effect of treatment 2 periods later

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- τ_S gives the marginal effect of treatment S periods later

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→ The cumulative effect q periods after treatment is:

$$T_q = \sum_{s=0}^{q} \tau_s$$

An example: Cyclones and economic growth

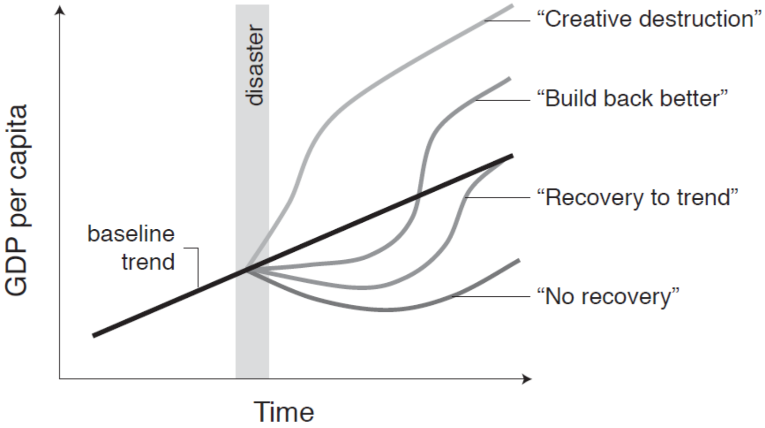
Policy issue:

- Climate change is projected to increase the intensity and number of cyclones
- What do cyclones actually do to an economy?
- How long do these effects last?

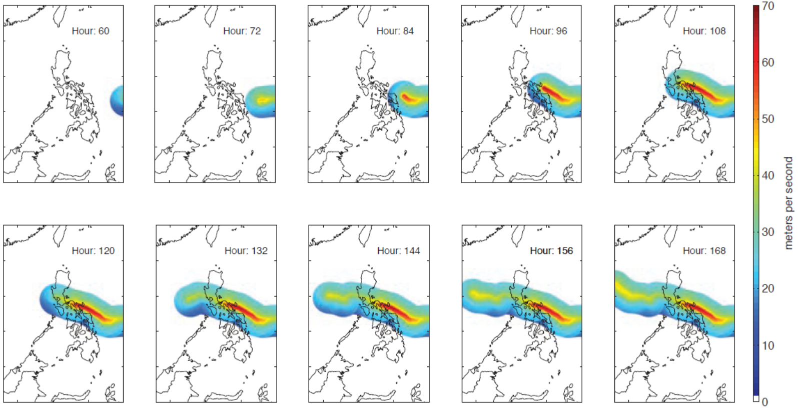
Approach:

- Construct a (cool, sciency) model of every hurricane 1950-2008
- Combine this with data on economic growth around the world
- Nobody randomized hurricanes
- ...but conditional on location and time FE, they are arguably exogenous
- Use a distributed lag model to compute cumulative effects

What are we trying to learn?



These hurricane data are so cool



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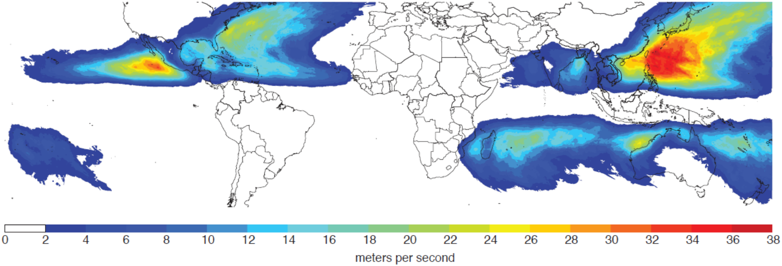


Estimating the effects of hurricanes on growth

The authors will run a version of:

$$Y_{it} = \sum_{s=0}^{S} \tau_s D_{i,t-s} + \alpha_i + \delta_t + \beta X_{it} + \varepsilon_{it}$$

where

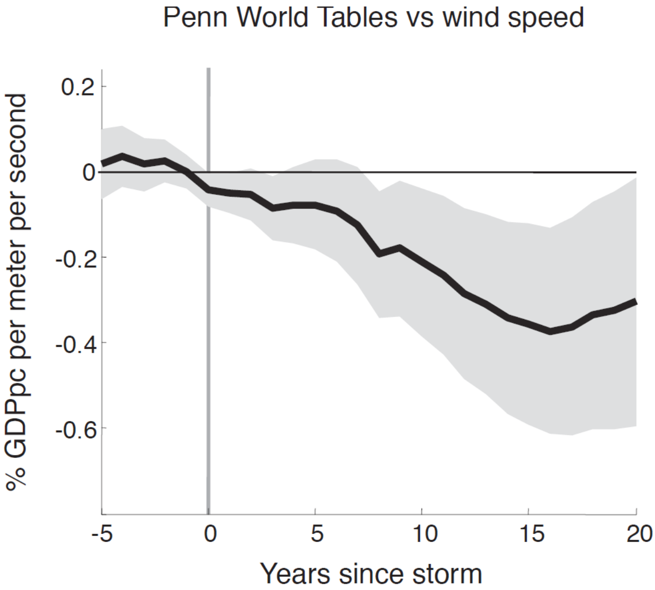
$Y_{it} = ln(GDP_{i,t}) - ln(GDP_{i,t-1})$ is the change in economic growth from $t-1$ to t

$D_{i,t}$ is the hurricane (scaled by windspeed)

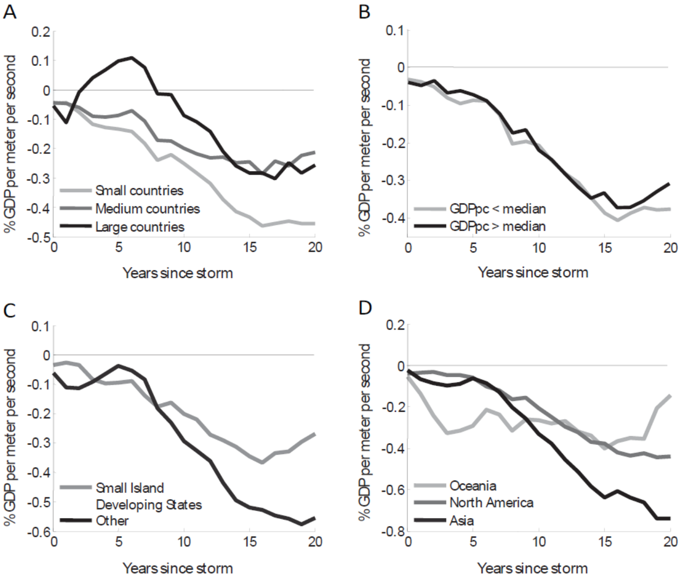
The unit of observation is the country-year

They then compute $T_q = \sum_{s=0}^{q} \tau_s$: the cumulative effects of a hurricane

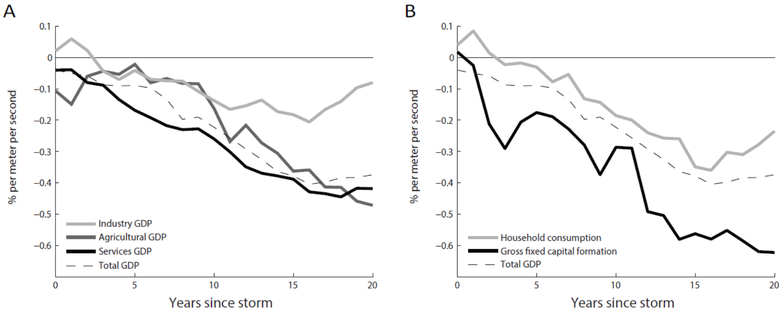
Main result


Penn World Tables vs wind speed
% GDPpc per meter per second
0.2
0
-0.2
-0.4
-0.6
-5
0
5
10
15
20
Years since storm

Results in different locations



Results for different sectors



How do these effects compare to other Bad Stuff?

Event Type	Effect on Income	Observed After	In-Sample Probability
Temperature increase (+1°C)*[1]	−1.0%	10 yrs	6.4%
Civil war[2]	−3.0%	10 yrs	6.3%
Tax increase (+1% GDP)**[3]	−3.1%	4 yrs	†16.8%
1 standard deviation cyclone	**−3.6%**	**20 yrs**	**14.4%**
Currency crisis[2]	−4.0%	10 yrs	34.7%
Weakening executive constraints [2]	−4.0%	10 yrs	3.7%
90th percentile cyclone	**−7.4%**	**20 yrs**	**5.8%**
Banking crisis[2]	−7.5%	10 yrs	15.7%
Financial crisis[4]	−9.0%	2 yrs	<0.1%
99th percentile cyclone	**−14.9%**	**20 yrs**	**0.6%**

*Poor countries only. **USA only. †Number of quarters with any tax change.

[1]Dell, Jones & Olken (AEJ: Macro, 2012), [2]Cerra & Saxena (AER, 2008), [3]Romer & Romer (AER, 2010), [4]Reinhart & Rogoff (AER, 2009)

Threats to identification

Recall our general fixed effects model:

$$Y_{it} = \tau D_{it} + \alpha_i + \delta_t + \beta X_{it} + \varepsilon_{it}$$

For the FE model to work, we need:

$$E[\varepsilon_{it} | \alpha_i, \delta_t, X_{it}] = 0$$

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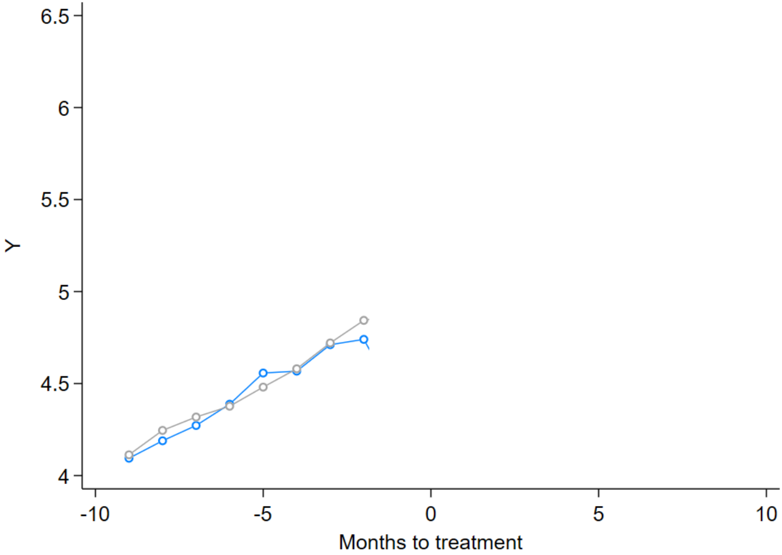
Threats to identification: Ashenfelter dip



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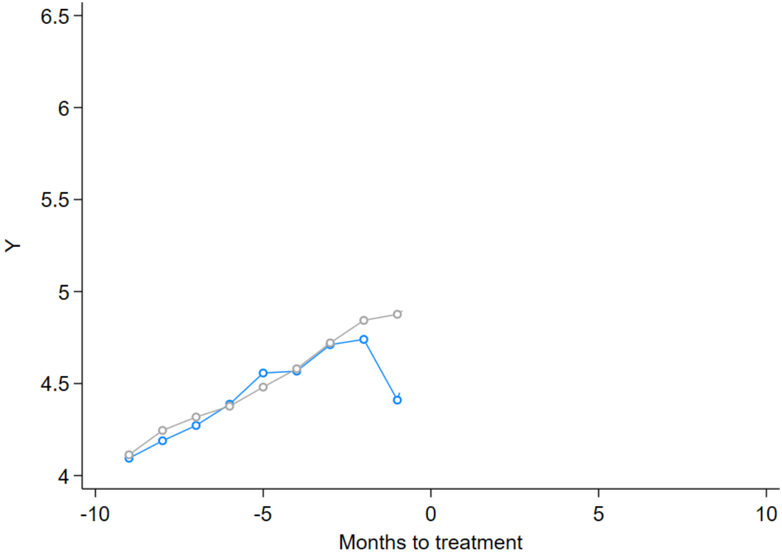

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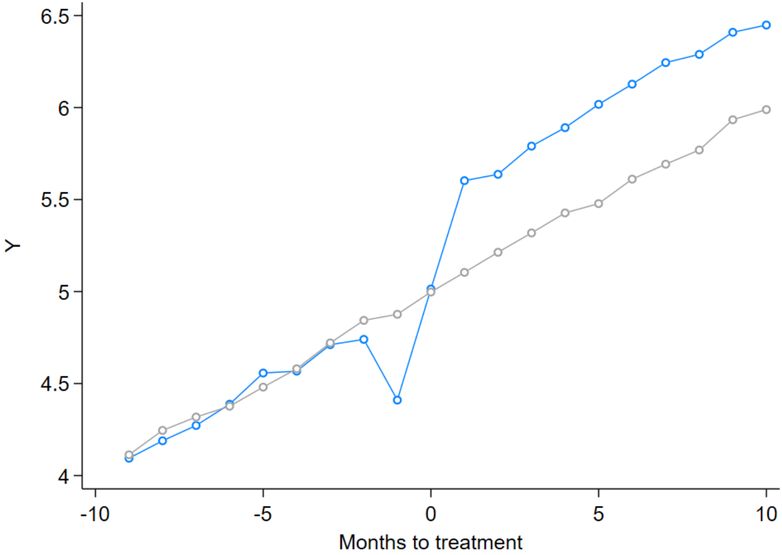

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$\rightarrow$ Clustered standard errors

  - These allow for correlations within cluster (eg, within individual)
  - Clustered SEs are usually (not always) bigger than OLS SEs
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  - You need at least $\approx$ 45 clusters for this to work

→ Typical panel data: cluster at the unit level

Even more differences

Differences-in-differences-in-differences (DDD)

Our standard DD estimator was:

$$\hat{\tau}^{DD} = (\bar{Y}(treat, post) - \bar{Y}(treat, pre)) - (\bar{Y}(untreat, post) - \bar{Y}(untreat, pre))$$

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We sometimes worry about:

- Non-parallel counterfactual trends
- Coincident treatments

Adding a third comparison can help with this:

- We use a comparison between affected and unaffected units or times

Differences-in-differences-in-differences (DDD)

Using a third difference can sometimes help with identification:

As an example, think about:

- Outcome: SO_x emissions
- Treatment: State-level regulations
- Timing: Pre/post
- *Additional grouping:* Industry type (some industries don't emit SO_x)

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→ What is the identifying assumption now?

→ The difference in trends over time between regulated and unregulated states is similar for emitters and non-emitters

Differences-in-differences-in-differences (DDD)

We can write this estimator formally as:

$$\hat{\tau}^{DDD} = [(\bar{Y}(treat, post, affected) - \bar{Y}(treat, pre, affected))$$
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This relaxes our identifying assumption:

- **DD:** Treated and untreated groups are on similar (counterfactual) trends

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This relaxes our identifying assumption:

- **DD:** Treated and untreated groups are on similar (counterfactual) trends
- **DDD:** Differences in trends between treated and untreated impact affected and unaffected groups similarly

Differences-in-differences-in-differences (DDD)

Affected group ($Affected_j = 1$)

	Post	Pre	Difference
Treated	a	b	$a - b$
Untreated	c	d	$c - d$
Difference	$a - c$	$b - d$	$(a - b) - (c - d)$

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Unaffected group ($Affected_j = 0$)

	Post	Pre	Difference
Treated	e	f	$e - f$
Untreated	g	h	$g - h$
Difference	$e - g$	$f - h$	$(e - f) - (g - h)$

Differences-in-differences-in-differences (DDD)

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Unaffected group ($Affected_j = 0$)

	Post	Pre	Difference
Treated	e	f	$e - f$
Untreated	g	h	$g - h$
Difference	$e - g$	$f - h$	$(e - f) - (g - h)$

DDD estimate is the differences between the two DID estimates:

$$[(a - b) - (c - d)] - [(e - f) - (g - h)].$$

Differences-in-differences-in-differences (DDD)

Just like the DD estimator, we can do the DDD estimator via regression:

$$
\begin{aligned}
Y_{ijt} = \beta_0 &+ \beta_1 Treat_i + \beta_2 Post_t + \beta_3 Affected_j + \beta_4 (Treat_i \times Post_t) \\
&+ \beta_5 (Post_t \times Affected_j) + \beta_6 (Treat_i \times Affected_j) \\
&+ \tau (Treat_i \times Post_t \times Affected_j) + \varepsilon_{ijt}
\end{aligned}
$$

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- You can do this with fixed effects too
- You just can't have a $Treat_i \times Post_t \times Affected_j$ FE

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For fun on your own: Show that τ actually gets $\hat{\tau}^{DDD}$

An example: The Clean Water Act

Policy issue:

- Rivers in the U.S. used to be incredibly polluted
- We might want to clean these rivers up
- Is this clean-up worth it?

Approach:

- Look at the Clean Water Act
- Estimate the effects of the CWA on water pollution
- We didn't randomize water treatment plants
- ...but we can do a pre-vs-post, treated-vs-untreated, upstream-vs-downstream comparison

$\rightarrow$ Use a DDD model to estimate treatment effects

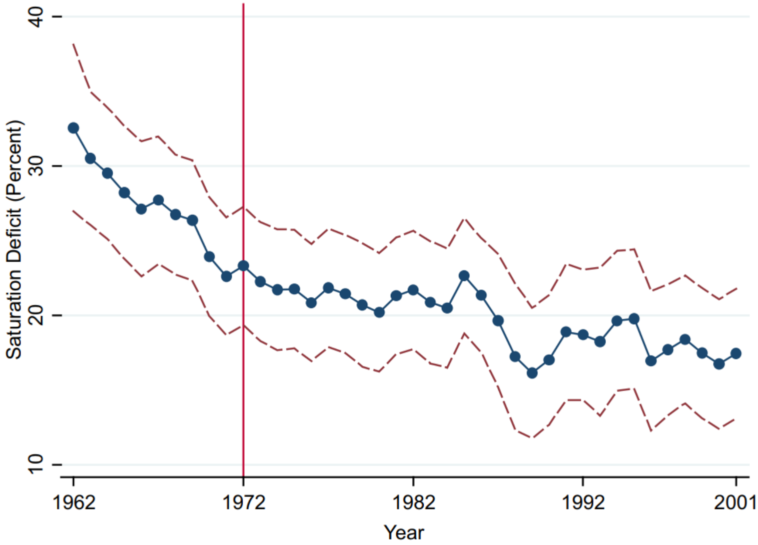
Estimating the effects of the Clean Water Act

The authors will run a version of:

$$\begin{aligned} Y_{ijt} = \beta_0 &+ \beta_1 Treat_i + \beta_2 Post_t + \beta_3 Downstream_j + \beta_4 (Treat_i \times Post_t) \\ &+ \beta_5 (Post_t \times Downstream_j) + \beta_6 (Treat_i \times Downstream_j) \\ &+ \tau (Treat_i \times Post_t \times Downstream_j) + \varepsilon_{ijt} \end{aligned}$$

where
Y_{ijt} is pollution at plant i in year y by downstream status j
$Treat_i \times Post_t \times Downstream_j$ turns on for plants after they've gotten a grant

Pollution over time



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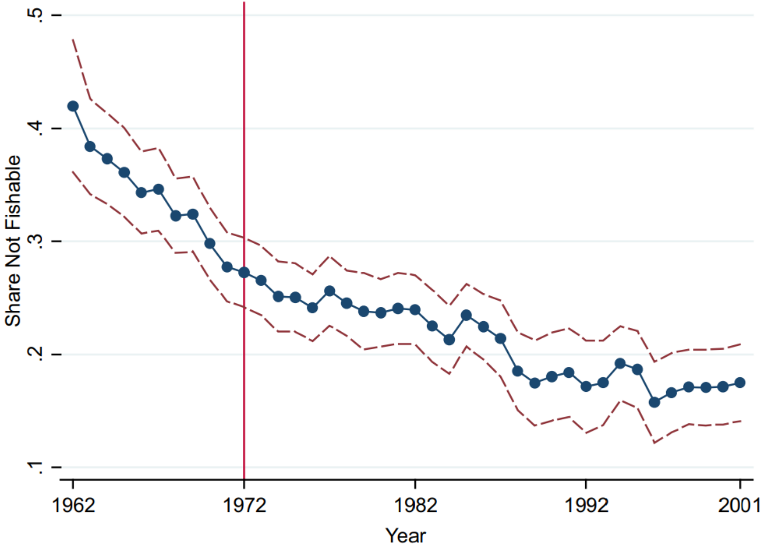
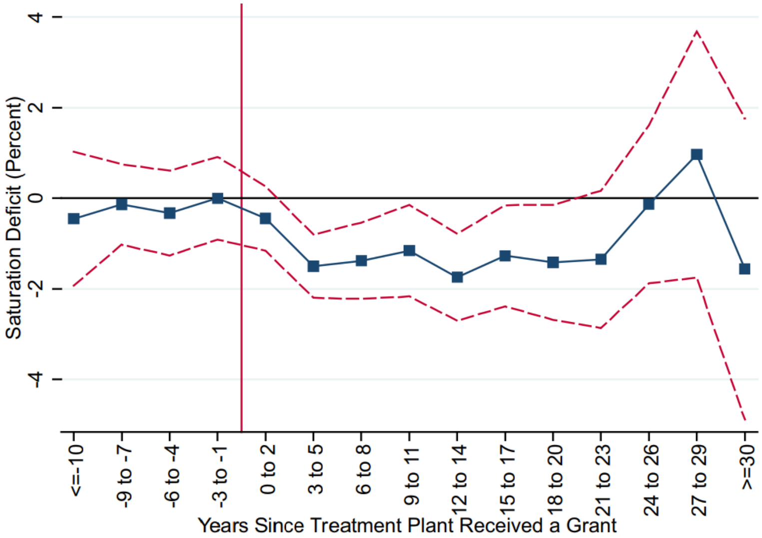


Causal effects of treatment



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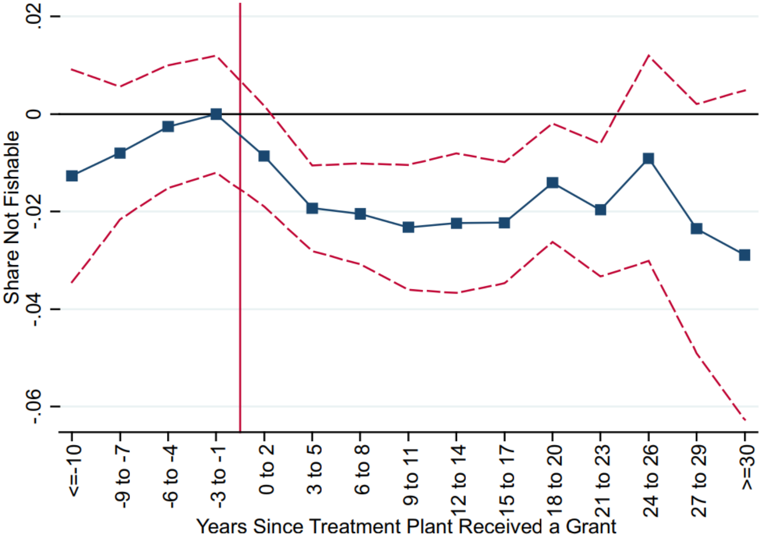


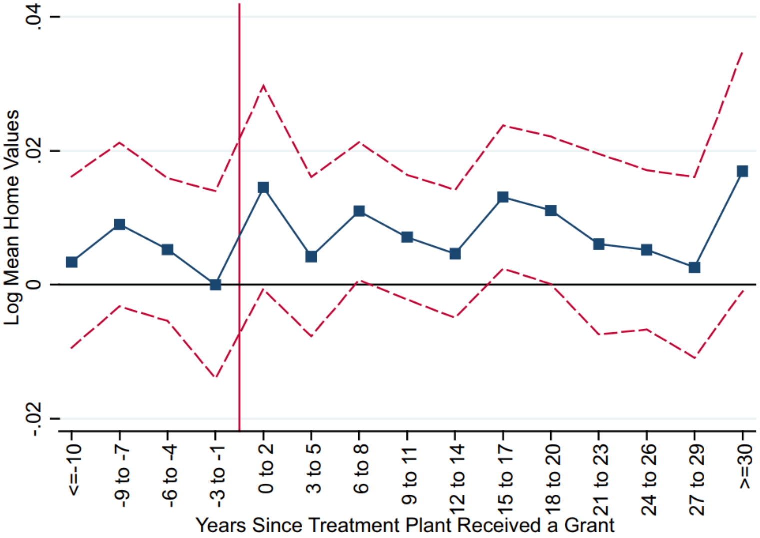
Causal effects of treatment

TABLE II
EFFECTS OF CLEAN WATER ACT GRANTS ON WATER POLLUTION

	Main pollution measures		Other pollution measures			
	Dissolved oxygen deficit (1)	Not fishable (2)	Biochemical oxygen demand (3)	Fecal coliforms (4)	Not swimmable (5)	Total suspended solids (6)
Downstream *Cumul. # grants	-0.681^{***}	-0.007^{**}	-0.104^{**}	-204.059^{**}	-0.004^{*}	-0.497
	(0.206)	(0.003)	(0.041)	(98.508)	(0.002)	(0.635)
N	55,950	60,400	28,932	34,550	60,400	30,604
Dep. var. mean	17.092	0.328	4.411	5,731.028	0.594	42.071
Fixed effects:						
Plant-downstream	Yes	Yes	Yes	Yes	Yes	Yes
Plant-year	Yes	Yes	Yes	Yes	Yes	Yes
Downst.-basin-year	Yes	Yes	Yes	Yes	Yes	Yes
Weather	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Each observation in a regression is a plant-downstream-year tuple. Data cover 1962–2001. Dissolved oxygen deficit equals 100 minus dissolved oxygen saturation, measured in percentage points. Dependent variable mean describes mean in 1962–1972. Standard errors are clustered by watershed. Asterisks denote p-value $< .10$ (*), $< .05$ (**), or $< .01$ (***).

Home values over time



Cost-effectiveness

Clean Water Act Grants: Costs and Effects on Home Values ($2014Bn)

	(1)	(2)	(3)	(4)
Ratio: Change in home	0.06	0.26	0.22	0.24
values/costs	(0.03)	(0.36)	(0.36)	(0.41)
p-value: ratio = 0	[0.05]	[0.46]	[0.55]	[0.56]
p-value: ratio = 1	[0.00]	[0.04]	[0.03]	[0.06]
Change in value of housing ($Bn)	15.92	89.25	73.7	91.97
Costs ($Bn)				
Capital: fed.	86.24	102.26	102.26	114.16
Capital: local	35.81	41.81	41.81	48.00
Variable	166.1	197.36	197.36	222.81
Total	288.15	341.44	341.44	384.97
Max distance homes to river (miles)	1	25	25	25
Include rental units			Yes	Yes
Include nonmetro areas				Yes

Recap

TL;DR:

1. We like the difference-in-differences approach a lot
2. We discussed estimation with fixed effects
3. And covered the event study and distributed lag versions

For next class

Topics:

- Regression discontinuity I

Reading: Ito (2015). You can skim:

- I: Conceptual framework